

logging is also one of the areas of failure for the Payment Card Industry's (PCI) Data Security Standard (DSS).

Logstash

Logstash is an open source tool for managing events and logs. It acts as a log analyser and runs on top of the centralised log management server. You may enable logging to track each and every system activity, especially on production servers, and you may do the same on all the production servers deployed in your infrastructure. So a huge amount of logs are generated and all are forwarded to the log server based on your configuration. Within the generated logs, tracing a particular event is a very tedious job for a systems admin. Logstash makes the systems admin's job a lot easier by parsing logs based on a requirement and generating a report. You can send logs from Rsyslog, Syslog and Syslog-ng server to the Logstash server for analysis. I would like to mention another log analyser tool called Octopussy, which is also an open source tool and worth a try.

Git

Git is an open source distributed version control system for small and large projects. This tool is used by both systems administrators and developers. Git has received a lot of attention in the open source world and I just want to make sure the tool is added to my list.

Puppet

Puppet is an open source configuration management tool. Most of the systems admin's tasks are repetitive and boring. So automating these is the only way to accomplish them quickly. Automation also helps in avoiding manual configuration errors. As I mentioned earlier, configuration is the most important task for systems administrators. I have selected Puppet among all other configuration management tools as it is one of the best adopted by many IT organisations and cloud solution providers. It helps systems administrators to deploy the custom configuration in a number of systems and maintain it throughout the life cycle.

OCS Inventory NG

OCS Inventory NG, which stands for Open Computers and Software Inventory Next Generation, is an open source solution for managing IT assets. Most of the time systems administrators run around to collect IT asset information reports when there is an IT audit. This tool helps them to discover all the devices connected to the network and to collect the inventory of software and hardware components installed on a system. It is also capable of deploying software and configuration scripts on a system.

RackTables

RackTables is an open source tool that is used as a data centre asset management system. Usually, data centre assets are tracked in spread sheets by the systems admin or

management team. Maintaining all the assets in a spread sheet and generating a report based on needs is a cumbersome task. RackTables is a tool that is handy when creating the data centre rack layout and for document network devices, servers and other components installed in a rack. It also helps to track IP address space. By looking at the rack layout, you can easily make out the remaining space left on it and generate reports.

Clonezilla

Clonezilla is open source software for disk imaging and cloning. This tool helps systems administrators in system deployment, bare metal backup and recovery. Cloning is one of the easiest ways to take backups of a particular partition or an entire disk, and it also helps to restore it. Two flavours of Clonezilla are available—Clonezilla Live and Clonezilla SE (server edition). Clonezilla Live is suitable for a single machine backup and restore. Clonezilla SE is for massive deployment. Clonezilla supports Linux, Windows and other operating systems, and a wide variety of file systems.

SystemRescueCD

Nowadays the bootable CD/DVD ISO of most Linux distributions has an option to boot the system into recovery/rescue mode in case of system failure. Though we have a rescue option available in CD/DVD ISO, SystemRescueCD brings more valuable tools to repair the corrupted system. SystemRescueCD is a Linux system rescue disk available as a bootable CD-ROM and USB stick. It's for administering or repairing your system and restoring data after a crash.

Ntopng

An important routine task of a systems admin is monitoring network traffic and keeping it controlled. Ntopng provides network traffic data of the active hosts and sorts the traffic according to different protocols. It is an open source tool and the next generation version of the original Ntop. It provides a nice Web interface to navigate all the information. 

References

- [1] Nmon (<http://sourceforge.net/projects/nmon/>)
- [2] Rsyslog (<http://sourceforge.net/projects/rsyslog/>)
- [3] Logstash (<http://logstash.net>)
- [4] Git (<http://git-scm.com>)
- [5] Puppet (<http://puppetlabs.com>)
- [6] OCS Inventory NG (<http://www.ocsinventory-ng.org/en/>)
- [7] RackTables (<http://racktables.org/>)
- [8] Clonezilla (<http://clonezilla.org/>)
- [9] SystemRescueCD (<http://sourceforge.net/projects/systemrescuecd/>)
- [10] Ntopng (<https://www.ntop.org/products/ntop/>)

By: Kannan K

The author works as a software development engineer at the Dell India R&D Centre in Bengaluru, and has eight years of experience in the Linux and virtualisation domains. He is interested in networking and security technologies.